Lesson notes · for you

Who a detector misses

University, vision 2

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Filters, then ReLU, then pooling
- Each part throws detail away

Activity: In the lab

1. Read the close-set test face's score and the line under the pictures.
2. Read the wide face's score: how far under the line is it?
3. Read the box's score and say why below the line is right for it.

Discussion

- Why is the training shape not a test?

Check yourself

- Name the group the detector missed.

Try it this week
State the line and the wide face's score from the lab.

Worksheet

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Name _____ Class _____ Date _____

1 **At one spot, a filter's response is −3. What does ReLU pass on to pooling?**

◯ -3 ◯ 0 ◯ 3

2 **A high score on the training shape is enough to trust a detector on new people.**

◯ True ◯ False

3 **Match each test picture to what the detector did.**

Close-set test face	• • • •	Below the line: correctly not a face
Wide face	• • • •	Above the line: called a face
Box	• • • •	Below the line: missed

4 **Order a fair check of a detector.**

☐ Test on shapes it did not train on

☐ Report who it misses

☐ Train on one set

✎ **Which mistake is worse here: missing the wide face, or calling the box a face?**

Check your answers

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1 **0**

ReLU keeps positive responses and turns negative ones into 0, so −3 becomes 0.

2 **False**

The wide face is still a face, and the detector missed it.

3 **Close-set test face — Above the line: called a face**
Wide face — Below the line: missed
Box — Below the line: correctly not a face

Same idea, wider spacing, lower score: 0.548 against a line of 0.55.

4 **Top to bottom: 2, 3, 1**

The miss is invisible if you only rescore the training pictures.